

Chapter 04 Part 2: Types of Reactions

1.

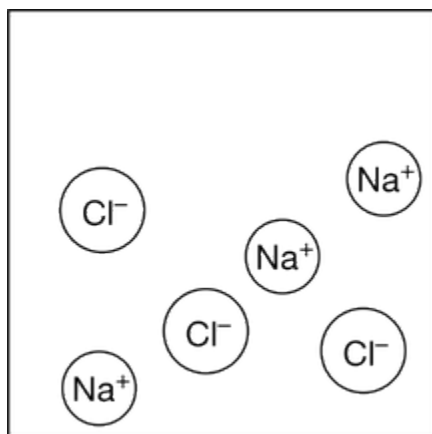


Diagram 1

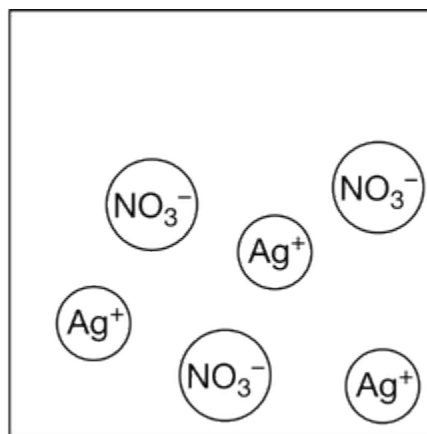
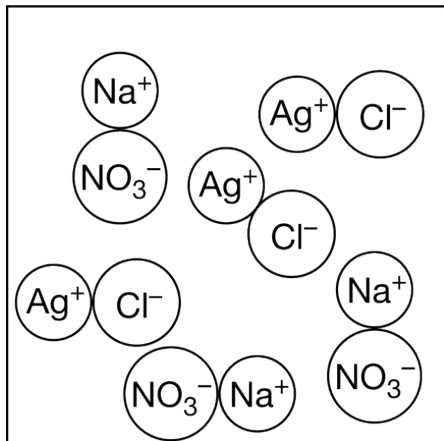


Diagram 2

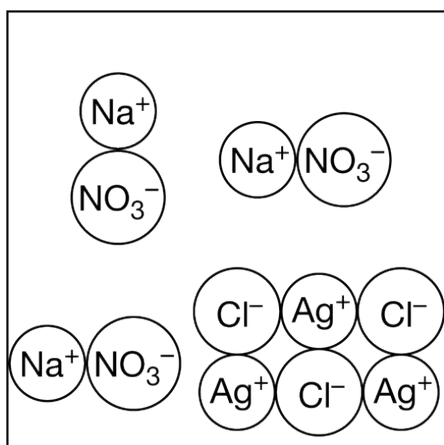
The diagrams above represent solutes present in two different dilute aqueous solutions before they were mixed. Water molecules are not shown. When the solutions were combined, a precipitation reaction took place. Which of the diagrams below is the best particle representation of the mixture after the precipitation reaction occurred?

Chapter 04 Part 2: Types of Reactions

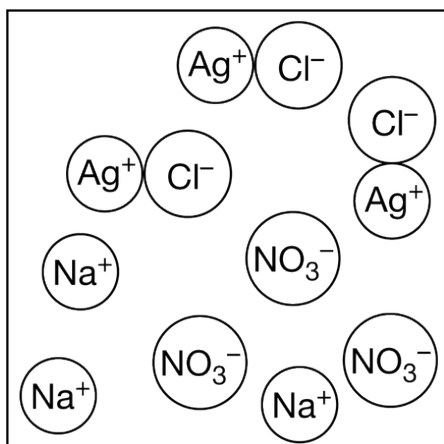
(A)



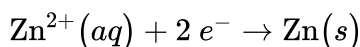
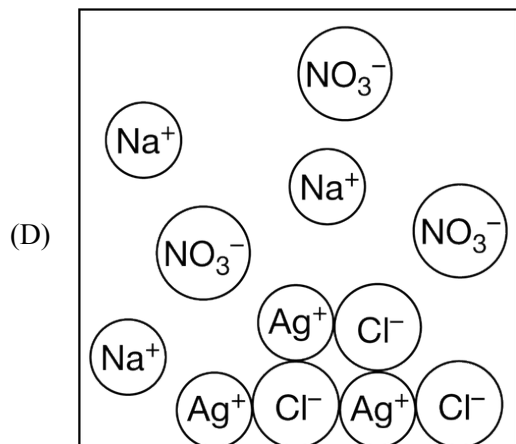
(B)



(C)

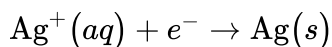
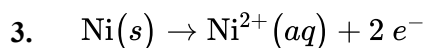


Chapter 04 Part 2: Types of Reactions



The half-reactions for the oxidation-reduction reaction between $\text{Al}(s)$ and $\text{Zn}^{2+}(aq)$ are represented above. Based on the half-reactions, what is the coefficient for $\text{Al}(s)$ if the equation for the oxidation-reduction reaction is balanced with the smallest whole-number coefficients?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

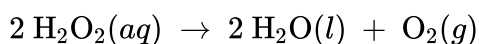


Which of the following is the balanced net ionic equation for an oxidation-reduction reaction between $\text{Ag}^{+}(aq)$ and $\text{Ni}(s)$ based on the half-reactions represented above?

- (A) $\text{Ag}^{+}(aq) + \text{Ni}(s) \rightarrow \text{Ag}(s) + \text{Ni}^{2+}(aq)$
- (B) $2 \text{Ag}^{+}(aq) + \text{Ni}(s) \rightarrow \text{Ag}(s) + 2 \text{Ni}^{2+}(aq)$
- (C) $\text{Ag}^{+}(aq) + 2 \text{Ni}(s) \rightarrow \text{Ag}(s) + 2 \text{Ni}^{2+}(aq)$
- (D) $2 \text{Ag}^{+}(aq) + \text{Ni}(s) \rightarrow 2 \text{Ag}(s) + \text{Ni}^{2+}(aq)$

Chapter 04 Part 2: Types of Reactions

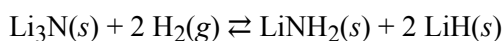
The following balanced equation represents the decomposition of hydrogen peroxide, H_2O_2 .



A student carries out an experiment in which a sample of $\text{H}_2\text{O}_2(aq)$ decomposes in the presence of $\text{MnO}_2(s)$. The student measures the volume of $\text{O}_2(g)$ produced by the reaction at regular intervals while the temperature is held constant. The student's data are given in the following table.

Time (s)	Volume of $\text{O}_2(g)$ Collected (mL)
0	0
60	42
120	64
180	76
240	82
300	86
360	88
420	89

4. The student claims that the reaction is an oxidation-reduction reaction. Which of the following statements correctly describes a change in oxidation number as the decomposition reaction proceeds?
- (A) The oxidation number of hydrogen changes from +2 to +1.
(B) The oxidation number of hydrogen changes from +1 to +2.
(C) The oxidation number of oxygen changes from -2 to -1.
(D) The oxidation number of oxygen changes from -1 to 0.
-

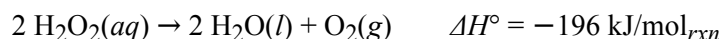


$$\Delta H^\circ = -192 \text{ kJ/mol}_{rxn}$$

Because pure H_2 is a hazardous substance, safer and more cost effective techniques to store it as a solid for shipping purposes have been developed. One such method is the reaction represented above, which occurs at 200°C .

5. Which of the following happens to H atoms in the forward reaction?
- (A) H atoms are oxidized only.
(B) H atoms are reduced only.
(C) H atoms are both oxidized and reduced.
(D) H atoms are neither oxidized nor reduced.
-

Chapter 04 Part 2: Types of Reactions

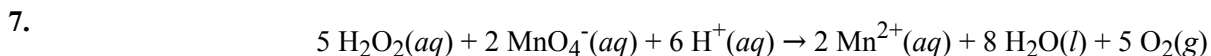


The decomposition of $\text{H}_2\text{O}_2(aq)$ is represented by the equation above. A student monitored the decomposition of a 1.0 L sample of $\text{H}_2\text{O}_2(aq)$ at a constant temperature of 300. K and recorded the concentration of H_2O_2 as a function of time. The results are given in the table below.

Time (s)	$[\text{H}_2\text{O}_2]$
0	2.7
200.	2.1
400.	1.7
600.	1.4

6. Which of the following identifies the element(s) being oxidized and reduced in the reaction?

- (A) Hydrogen is oxidized and oxygen is reduced.
 - (B) Oxygen is oxidized and hydrogen is reduced.
 - (C) Oxygen is both oxidized and reduced.
 - (D) No elements are oxidized or reduced; the reaction is not a redox reaction.
-

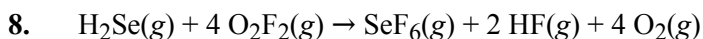


In a titration experiment, $\text{H}_2\text{O}_2(aq)$ reacts with aqueous $\text{MnO}_4^-(aq)$ as represented by the equation above. The dark purple KMnO_4 solution is added from a buret to a colorless, acidified solution of $\text{H}_2\text{O}_2(aq)$ in an Erlenmeyer flask. (Note: At the end point of the titration, the solution is a pale pink color.)

Which element is being oxidized during the titration, and what is the element's change in oxidation number?

Chapter 04 Part 2: Types of Reactions

- (A) Oxygen, which changes from -1 to 0
- (B) Oxygen, which changes from 0 to -2
- (C) Manganese, which changes from -1 to +2
- (D) Manganese, which changes from +7 to +2



Which of the following is true regarding the reaction represented above?

- (A) The oxidation number of O does not change.
- (B) The oxidation number of H changes from -1 to +1.
- (C) The oxidation number of F changes from +1 to -1.
- (D) The oxidation number of Se changes from -2 to +6.
- (E) It is a disproportionation reaction for F.

9. Which of the following represents a process in which a species is reduced?

- (A) $\text{Ca(s)} \rightarrow \text{Ca}^{2+}\text{(aq)}$
- (B) $\text{Hg(l)} \rightarrow \text{Hg}_2^{2+}\text{(aq)}$
- (C) $\text{Fe}^{2+}\text{(aq)} \rightarrow \text{Fe}^{3+}\text{(aq)}$
- (D) $\text{NO}_3^-\text{(aq)} \rightarrow \text{NO(g)}$
- (E) $\text{SO}_3^{2-}\text{(aq)} \rightarrow \text{SO}_4^{2-}\text{(aq)}$

The following questions refer to the below.

M^+ is an unknown metal cation with a +1 charge. A student dissolves the chloride of the unknown metal, MCl , in enough water to make 100.0 mL of solution. The student then mixes the solution with excess AgNO_3 solution, causing AgCl to precipitate. The student collects the precipitate by filtration, dries it, and records the data shown below. (The molar mass of AgCl is 143 g/mol.)

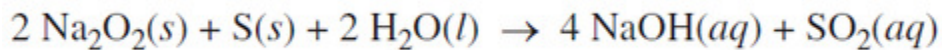
Mass of unknown chloride, MCl	0.74 g
Mass of filter paper	0.80 g
Mass of filter paper plus AgCl precipitate	2.23 g

10. During the course of the experiment, which of the following happens to the NO_3^- ions?

- (A) They are oxidized by Cl^- ions.
 - (B) They are reduced to NO_2^- ions.
 - (C) They are decomposed by reacting with M^+ ions.
 - (D) They remain dissolved in the filtrate solution.
-

Chapter 04 Part 2: Types of Reactions

When water is added to a mixture of $\text{Na}_2\text{O}_2(s)$ and $\text{S}(s)$, a redox reaction occurs, as represented by the equation below.



$$\Delta H_{298}^\circ = -610 \text{ kJ/mol}_{\text{rxn}}; \Delta S_{298}^\circ = -7.3 \text{ J/(K}\cdot\text{mol}_{\text{rxn}})$$

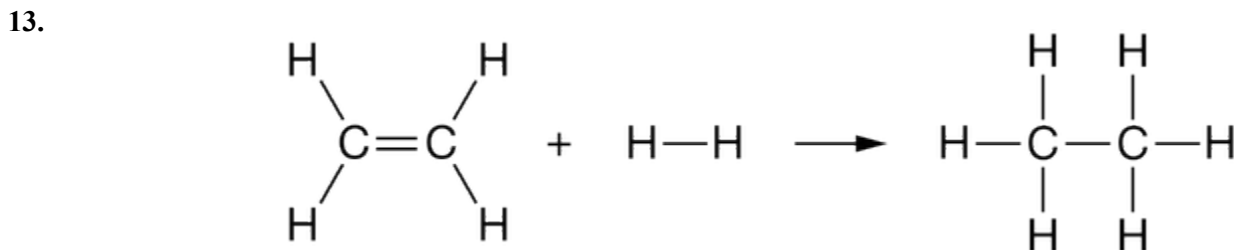
11. Atoms of which element are reduced in the reaction?

- (A) S; each atom loses four electrons
 - (B) Na in Na_2O_2 ; each atom loses one electron
 - (C) O in Na_2O_2 ; each atom gains one electron
 - (D) O in H_2O ; each atom gains one electron
-

12.
$$2 \text{MnO}_4^-(aq) + 10 \text{Br}^-(aq) + 16 \text{H}^+(aq) \rightarrow 2 \text{Mn}^{2+}(aq) + 5 \text{Br}_2(aq) + 8 \text{H}_2\text{O}(l)$$

How many electrons are transferred in the reaction represented by the balanced equation above?

- (A) 2
- (B) 4
- (C) 5
- (D) 8
- (E) 10



When $\text{C}_2\text{H}_4(g)$ reacts with $\text{H}_2(g)$, the compound $\text{C}_2\text{H}_6(g)$ is produced, as represented by the equation above. The reaction is correctly classified as which of the following types?

- (A) Acid-base, because two hydrogen atoms are donated to $\text{C}_2\text{H}_4(g)$.
 - (B) Precipitation, because two reactant species form a single product.
 - (C) Decomposition, because pure $\text{H}_2(g)$ is consumed.
 - (D) Oxidation-reduction, because $\text{H}_2(g)$ is oxidized.
-

Chapter 04 Part 2: Types of Reactions

These questions refer to the following compounds at 25°C and 1.0 atm.

- (A) CO_2
- (B) PbO_2
- (C) CaO
- (D) N_2O_5
- (E) Cu_2O

14. Which of the following compounds at 25°C and 1.0 atm contains an element in a +1 oxidation state?

- (A) CO_2
 - (B) PbO_2
 - (C) CaO
 - (D) N_2O_5
 - (E) Cu_2O
-

Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits each statement. A choice may be used once, more than once, or not at all in each set.

The following questions refer to the reactions represented below.

- (A) $\text{H}_2\text{SeO}_4(aq) + 2 \text{Cl}^-(aq) + 2 \text{H}^+(aq) \rightarrow \text{H}_2\text{SeO}_3(aq) + \text{Cl}_2(g) + \text{H}_2\text{O}(l)$
- (B) $\text{S}_8(s) + 8 \text{O}_2(g) \rightarrow 8 \text{SO}_2(g)$
- (C) $3 \text{Br}_2(aq) + 6 \text{OH}^-(aq) \rightarrow 5 \text{Br}^-(aq) + \text{BrO}_3^-(aq) + 3 \text{H}_2\text{O}(l)$
- (D) $\text{Ca}^{2+}(aq) + \text{SO}_4^{2-}(aq) \rightarrow \text{CaSO}_4(s)$
- (E) $\text{PtCl}_4(s) + 2 \text{Cl}^-(aq) \rightarrow \text{PtCl}_6^{2-}(aq)$

15. A precipitation reaction

- (A) $\text{H}_2\text{SeO}_4(aq) + 2 \text{Cl}^-(aq) + 2 \text{H}^+(aq) \rightarrow \text{H}_2\text{SeO}_3(aq) + \text{Cl}_2(g) + \text{H}_2\text{O}(l)$
- (B) $\text{S}_8(s) + 8 \text{O}_2(g) \rightarrow 8 \text{SO}_2(g)$
- (C) $3 \text{Br}_2(aq) + 6 \text{OH}^-(aq) \rightarrow 5 \text{Br}^-(aq) + \text{BrO}_3^-(aq) + 3 \text{H}_2\text{O}(l)$
- (D) $\text{Ca}^{2+}(aq) + \text{SO}_4^{2-}(aq) \rightarrow \text{CaSO}_4(s)$
- (E) $\text{PtCl}_4(s) + 2 \text{Cl}^-(aq) \rightarrow \text{PtCl}_6^{2-}(aq)$

16. A reaction in which the same reactant undergoes both oxidation and reduction

Chapter 04 Part 2: Types of Reactions

- (A) $\text{H}_2\text{SeO}_4(aq) + 2 \text{Cl}^-(aq) + 2 \text{H}^+(aq) \rightarrow \text{H}_2\text{SeO}_3(aq) + \text{Cl}_2(g) + \text{H}_2\text{O}(l)$
- (B) $\text{S}_8(s) + 8 \text{O}_2(g) \rightarrow 8 \text{SO}_2(g)$
- (C) $3 \text{Br}_2(aq) + 6 \text{OH}^-(aq) \rightarrow 5 \text{Br}^-(aq) + \text{BrO}_3^-(aq) + 3 \text{H}_2\text{O}(l)$
- (D) $\text{Ca}^{2+}(aq) + \text{SO}_4^{2-}(aq) \rightarrow \text{CaSO}_4(s)$
- (E) $\text{PtCl}_4(s) + 2 \text{Cl}^-(aq) \rightarrow \text{PtCl}_6^{2-}(aq)$
-

17. A precipitation reaction

- (A) $2 \text{Mg}(s) + \text{O}_2(g) \rightarrow 2 \text{MgO}(s)$
- (B) $\text{Pb}^{2+}(aq) + \text{CrO}_4^{2-}(aq) \rightarrow \text{PbCrO}_4(s)$
- (C) $\text{SO}_3(g) + 2 \text{H}_2\text{O}(l) \rightarrow \text{H}_3\text{O}^+(aq) + \text{HSO}_4^-(aq)$
- (D) $2 \text{H}_2\text{O}(g) \rightarrow 2 \text{H}_2(g) + \text{O}_2(g)$
- (E) $\text{Ag}^+(aq) + 2 \text{NH}_3(aq) \rightarrow [\text{Ag}(\text{NH}_3)_2]^+(aq)$

18. An oxidation-reduction reaction that is also a synthesis reaction

- (A) $2 \text{Mg}(s) + \text{O}_2(g) \rightarrow 2 \text{MgO}(s)$
- (B) $\text{Pb}^{2+}(aq) + \text{CrO}_4^{2-}(aq) \rightarrow \text{PbCrO}_4(s)$
- (C) $\text{SO}_3(g) + 2 \text{H}_2\text{O}(l) \rightarrow \text{H}_3\text{O}^+(aq) + \text{HSO}_4^-(aq)$
- (D) $2 \text{H}_2\text{O}(g) \rightarrow 2 \text{H}_2(g) + \text{O}_2(g)$
- (E) $\text{Ag}^+(aq) + 2 \text{NH}_3(aq) \rightarrow [\text{Ag}(\text{NH}_3)_2]^+(aq)$

19. $\text{Cu}(s) + 2 \text{AgNO}_3(aq) \rightarrow \text{Cu}(\text{NO}_3)_2(aq) + 2 \text{Ag}(s)$

The reaction between solid copper and aqueous silver nitrate produces solid silver and a blue solution, as represented by the balanced equation shown above. Based on the balanced equation, which of the following identifies the oxidation and reduction half-reactions?

Chapter 04 Part 2: Types of Reactions

(A)

Oxidation Half-reaction	Reduction Half-reaction
$\text{Cu}(s) \rightarrow \text{Cu}^+(aq) + e^-$	$\text{Ag}^{2+}(aq) + 2 e^- \rightarrow \text{Ag}(s)$

(B)

Oxidation Half-reaction	Reduction Half-reaction
$\text{Cu}(s) \rightarrow \text{Cu}^{2+}(aq) + 2 e^-$	$\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$

(C)

Oxidation Half-reaction	Reduction Half-reaction
$\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$	$\text{Cu}(s) \rightarrow \text{Cu}^{2+}(aq) + 2 e^-$

(D)

Oxidation Half-reaction	Reduction Half-reaction
$\text{Ag}^{2+}(aq) + 2 e^- \rightarrow \text{Ag}(s)$	$\text{Cu}(s) \rightarrow \text{Cu}^+(aq) + e^-$

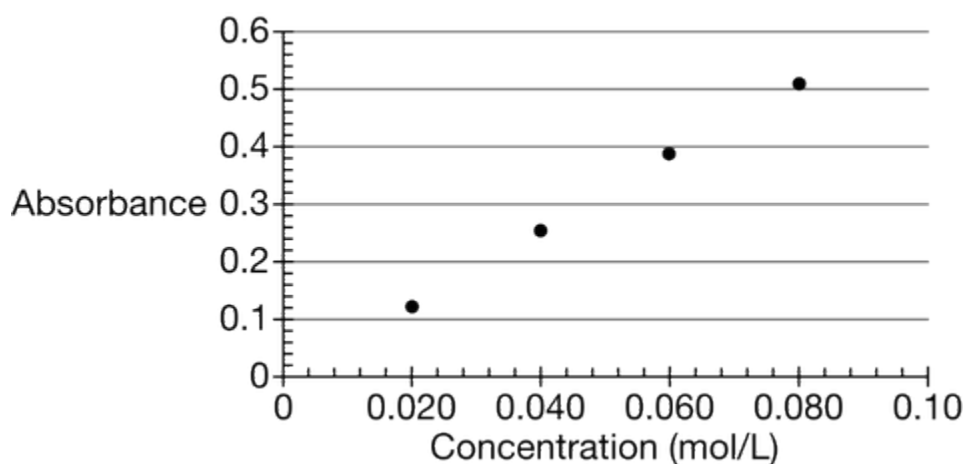
20. In which of the following species does sulfur have the same oxidation number as it does in H_2SO_4 ?

- (A) H_2SO_3
- (B) $\text{S}_2\text{O}_3^{2-}$
- (C) S^{2-}
- (D) S_8
- (E) SO_2Cl_2

Chapter 04 Part 2: Types of Reactions

An experiment was performed to investigate the reaction between Zn metal and $\text{Ni}^{2+}(aq)$ at different concentrations. Because the $\text{Ni}^{2+}(aq)$ ion is green, the extent of the reaction was determined using spectrophotometric analysis. Four 20.0 mL standard solutions of $\text{Ni}^{2+}(aq)$ were prepared by dissolving $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ (molar mass 240 g/mol) in water. The absorbance of each solution was measured; the results are shown both in the table below and in the following plot of the absorbance data.

Solution	$[\text{Ni}^{2+}]$	Absorbance
1	0.020	0.12
2	0.040	0.25
3	0.060	0.39
4	0.080	0.51



21. Which species is being oxidized in the reaction between Zn metal and $\text{NiCl}_2(aq)$?

- (A) Zn
- (B) Zn^{2+}
- (C) Ni
- (D) Ni^{2+}

22. $\text{I}_2(aq) + \text{C}_6\text{H}_8\text{O}_6(aq) \rightarrow \text{C}_6\text{H}_6\text{O}_6(aq) + 2\text{I}^-(aq) + 2\text{H}^+(aq)$

The compound $\text{C}_6\text{H}_8\text{O}_6$ reacts with I_2 according to the reaction represented by the equation above. The reaction is correctly classified as which of the following types?

- (A) Acid-base, because H^+ ions are produced.
- (B) Precipitation, because there are only two reactants but there are three products.
- (C) Double replacement, because both H and I appear as ions in the products.
- (D) Oxidation-reduction, because I_2 is reduced.

Chapter 04 Part 2: Types of Reactions

23. $\text{Zn}(s)$ is used to reduce other compounds in chemical reactions. If a chemist needs a substance that is more effective in its reducing ability, which of the following species would be the best choice?
- (A) Na
 - (B) H^+
 - (C) K^+
 - (D) Cl^-